StudyLadder: Project Plan (Women in Big Data Buildathon 2026)

Oct 18, 2026 · @Hansika Reddy Bandela

Vision

StudyLadder (working name) is an AI study-planning platform that finds the cheapest way for a student to pass their next exam: free campus help first, study groups second, paid tutors only when they are actually needed.

The problem. Students search for help right before exams, don't know their campus already offers free tutoring, and default to paid marketplaces. Campus centers, meanwhile, have no view of when demand will spike.

What makes it different. Most tutor marketplaces start from a tutor search, and their revenue grows with paid sessions. StudyLadder starts from the student's exam date and goal, and its agent is rewarded for saving the student money.

Typical tutor marketplace	StudyLadder
Starts with a tutor search	Starts with course + exam date
Paid options first	Free campus help first, paid last
Static profiles	AI plan, AI matching, AI quizzes inside the session
No view of demand	Anonymized demand data shared with campus centers and tutors
Generic audience	Built for under-resourced and first-gen students, with a women-in-STEM mentor option

Five pillars of the project

1. **Free-First Ladder:** the agent always checks free campus help, then study groups, before paid tutors.
2. **Exam Prep Planner:** an agent turns an exam date into a session-by-session plan.
3. **Live AI quiz:** online tutors generate and issue a quiz to the student inside the session.

4. **Demand intelligence:** aggregated exam dates reveal demand spikes by course and week.
5. **Equity focus:** a visible savings metric and a mentor tag for women in STEM.

Who it's for

Three user groups, each with one clear job the product does for them.

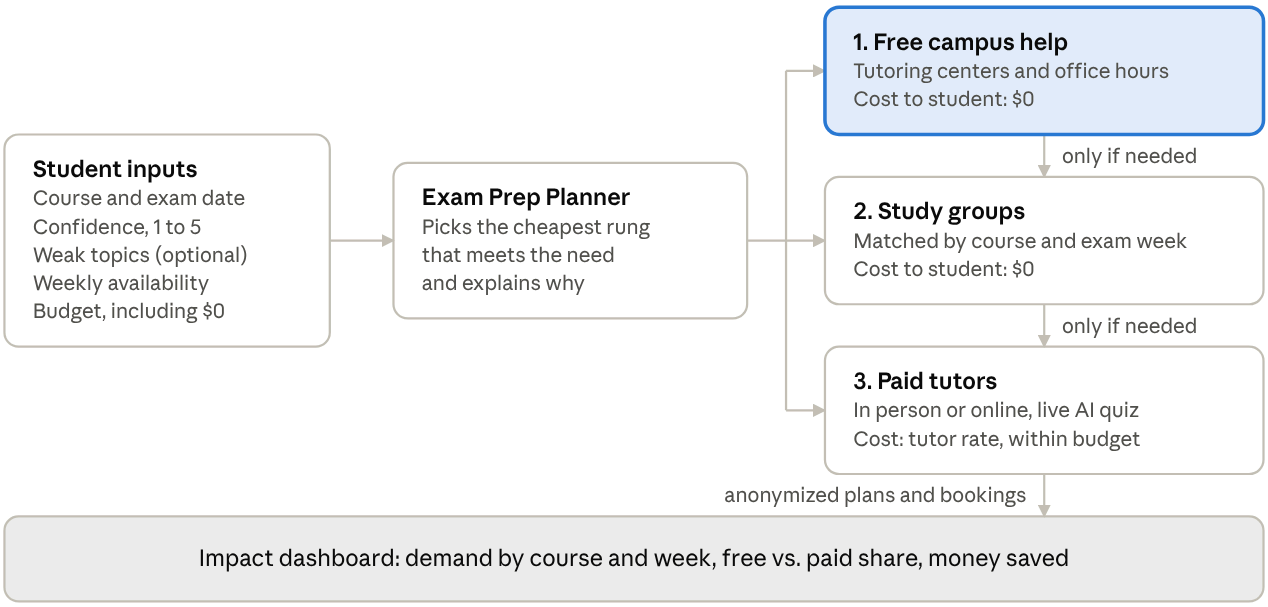
User	Who they are	Top pain point	What StudyLadder does
Student	Undergrad with an upcoming exam, often on a tight budget	Doesn't know what help exists or how much prep is enough	Builds a prep plan and finds the cheapest fit
Tutor (paid, in person or online)	Grad student, professional or former TA	Hard to be found by the right students, hard to prove credentials	Rich profile, matched leads, built-in AI quiz tool
Campus service	Tutoring center, writing lab, supplemental instruction	Can't see demand coming, underused services	Free listing plus a demand forecast by course and week

Primary persona for the demo: a second-year student, low confidence in Calculus II, exam in 16 days, budget of $0 to $30 per session, free Tuesday and Thursday evenings. Every demo screen should serve her.

Product at a glance

Students never start with a tutor list. The Exam Prep Planner takes their exam details and walks them up the ladder only as far as needed.

The agent starts at the free rung and climbs only as far as the student needs



Free-First Ladder · inputs, planner agent, 3 rungs, dashboard

Read left to right: inputs go to the planner, the planner picks the cheapest rung that meets the need, and every plan and booking feeds the dashboard as anonymized counts.

Features by priority

Ship the MVP table completely before touching Stretch. A polished MVP beats a broad, half-working app.

MVP (must ship by week 6)

Feature	For	What it does
University + location search	Student	Two result tabs: Free on-campus and Paid tutors
Filters	Student	Subject, course code, price per session, availability, qualifications, rating, in-person or online
Tutor profile	Tutor	Qualifications, current job or past experience, rate, weekly availability, courses taken with grade

Feature	For	What it does
Campus service listing	Campus	Hours, location, subjects covered, drop-in or appointment
Ratings and reviews	Student	Only students with a completed booking can review
Exam Prep Planner agent	Student	Course + exam date + confidence in, session plan out
Tutor Matcher agent	Student	Plain-language request in, top 3 matches with reasons out
Study group finder	Student	Create or join by course, exam date, time, place
Online session room with AI quiz	Tutor + student	Tutor generates a quiz from topics and issues it live
Impact dashboard	Judges, campus	Demand by course and week, money saved by free-first choices

Stretch (weeks 7 to 8, only if MVP is solid)

Feature	Why it helps
Last-minute help (next 24 hours) filter	Most searches happen right before exams
Group sessions with cost splitting	Lowers price per student
Session recap agent	Summary plus practice questions after each session
Review summarizer	One-paragraph "what students say" per tutor
Accessibility and language filters	Inclusion for students with learning differences and non-native English speakers
Women-in-STEM mentor tag	Fits the buildathon theme

Roadmap (pitch slide only, do not build)

- Real payments and payout
- Automated credential verification

- Scraping campus sites to scale listings to hundreds of universities
- Calendar sync and push reminders
- Forecasting model for demand by course

Seed data decision: 3 to 5 universities with real public tutoring info, 15 to 25 tutor profiles (some invented and labeled as sample data), and 10 to 15 study groups.

The AI agents

Four agents share one rule: recommend the cheapest option that meets the student's need, and always explain why.

1. Exam Prep Planner (signature feature, build first)

- **Inputs:** course, exam date, confidence (1 to 5), weak topics (optional), weekly availability, budget (including $0), preference for one-on-one or group.
- **Output:** a recommended number of sessions, a date for each, and the best resource type for each slot.
- **Starting heuristic** (state openly that this is a starting plan, not a guarantee):

Time to exam	Confidence 1 to 2	Confidence 3	Confidence 4 to 5
3+ weeks	1 session per week (3 to 4 total)	1 session every 10 days	1 review session in the last week
1 to 3 weeks	2 sessions per week	1 session per week	1 review session
Under 1 week	2 to 3 intensive sessions	1 to 2 sessions	1 quick check-in

- **Budget cap:** the plan never exceeds the student's budget. The agent fills slots in this order: free campus help, study groups, then paid tutors.
- **Last session** is always 2 days before the exam, so there is time to practice what was covered.
- **Implementation tip:** keep the heuristic as a small rules function in the app, and let the AI model write the explanation and adjust for the weak topics. That keeps results consistent and explainable.

2. Tutor Matcher

- Input is plain language, such as "failing Calc II, exam in two weeks, I learn from worked examples, under $30."
- The agent extracts course, urgency, learning style and budget, filters the database, and returns the top 3 with a one-line reason each.
- Matching signals: took this exact course, grade earned, rating, price fit, availability overlap, online or in person.

3. Live Quiz Agent (inside online sessions)

Covered in its own section below.

4. Session Recap Agent (stretch)

After a session it summarizes what was covered, lists weak spots seen in the quiz results, and generates 5 practice questions. Feeds the next session's plan.

Agent guardrails

- Every recommendation shows its reasoning in plain language.
- The agent only recommends tutors and services that exist in the database, never invented ones.
- Students see the free option even when a paid tutor scores higher on fit, labeled clearly.

Online tutoring and the live AI quiz

The live quiz is your most memorable demo moment: the tutor generates a quiz in seconds and the student answers it during the session.

How a session works

1. The student books an online slot with a tutor (or joins a free campus session).
2. The tutor opens the session page, which shows the student's course, exam date, weak topics and plan from the Exam Prep Planner.
3. The tutor clicks **Generate quiz**, picks topics, difficulty and number of questions (3 to 10), and optionally pastes lecture notes for the AI to base questions on.
4. The Live Quiz Agent drafts questions with answer keys and short explanations. The tutor can edit, delete or regenerate any question.

5. The tutor clicks **Issue to student**. The student sees the quiz on their side of the session page and answers in real time.
6. Results appear instantly for both. The agent flags weak topics and the tutor continues the session where it matters most.
7. After the session, scores are saved to the student's progress and feed the next plan (and the recap agent, if built).

Video: do not build video from scratch. Each session stores a meeting link (Zoom, Google Meet or Jitsi) and the quiz panel sits beside it. This saves weeks.

Quiz question types for the MVP: multiple choice and short numeric or text answer. Skip file uploads and free-form essays for now.

Why it is unique: the AI works for the tutor as a teaching assistant, so tutors with a StudyLadder profile deliver a better session. That is a reason for tutors to join that other platforms don't offer.

Quality controls

- Tutor always reviews questions before issuing, so the human stays in charge.
- Every question shows its source topic so wrong answers can be traced.
- A "report a bad question" button feeds a quality log.

Study groups

Study groups are the free step between campus tutoring and paid tutors, and they work for students with a $0 budget.

- **Create a group:** course, exam date, topic focus (for example "Chapter 5 practice problems"), meeting time, place (library room or online link), max size (default 6).
- **Join:** one-click RSVP with a seat counter. Waitlist if full.
- **Agent matching:** the Exam Prep Planner suggests groups that share the student's course, exam week and availability, and says why.
- **Group quiz (nice touch):** any member can ask the Quiz Agent for a 5-question group warm-up from the group's topic.
- **Optional group types:** women-in-STEM friendly, quiet study, first-gen students.

Safety basics

- Campus email verification before joining or creating a group.
- Public meeting places suggested by default for in-person groups.

- Report button and a visible code of conduct.
- No private contact details shown; chat stays inside the app.

The Big Data angle

The data story is what turns StudyLadder from a marketplace into a Big Data project: every exam date, booking and quiz result becomes an anonymized signal that helps students, tutors and campuses.

Data sources

Source	Use	Note
Platform events (searches, plans, bookings, quiz scores)	Demand and outcome analytics	Anonymized and aggregated; sample data for the demo
Public campus tutoring pages	Seed the free-services directory	Entered manually for 3 to 5 campuses
Public education data such as NCES/IPEDS	Context on enrollment and costs	Verify the exact tables before citing
Public wage data such as BLS tutor wages	Benchmark paid tutor rates	Verify the exact series before citing
Simulated exam-season dataset	Fill the dashboard realistically	Label clearly as simulated in the pitch

Dashboard views (build 4, not 10)

1. **Demand heatmap:** students with an exam in each course by week.
2. **Free vs. paid usage:** share of bookings that went to free campus help, groups and paid tutors.
3. **Money saved:** estimated dollars saved by following the free-first plan.
4. **Capacity gap:** weeks where demand for a course exceeds free tutoring slots.

Money-saved metric (define it simply and show the formula in the pitch):

```
\text{Saved} = \sum_{\text{sessions}} \left( \text{median paid rate for subject} - \text{actual cost} \right)
```

Analytics you can show from your DataCamp work

- Cleaning and joining the seed datasets (pandas or SQL).
- Price distribution by subject, with outliers handled.
- Demand by week relative to exam dates.
- A simple forecast of next week's demand per course (moving average is enough; call it a baseline).

Privacy: individual exam dates and quiz scores stay private. Dashboards show only aggregates, and hide any group smaller than 5 students.

Data model

Twelve entities cover the whole MVP. Create them in Base44 in this order, then seed them. Fields beyond these stay optional until the MVP is stable.

Entity	Key fields
University	name, city, state, latitude, longitude
Course	university, code (for example MATH 251), title, subject
Student	name, campus email, university, role (student or tutor), verified
TutorProfile	user, bio, degrees, certifications, current job, past experience, subjects, courses taken with grade, hourly rate, sliding-scale flag, online or in person, mentor tag, verified badge
Availability	tutor or service, weekday, start time, end time
CampusService	university, name, type (tutoring center, writing lab, supplemental instruction, office hours), subjects, hours, location, drop-in or appointment, link
Booking	student, tutor or service, session time, mode (online or in person), price, status, meeting link
Review	booking, rating (1 to 5), comment, created date
StudyGroup	course, exam date, topic focus, meeting time, place, max size, members, group type

Entity	Key fields
ExamPlan	student, course, exam date, confidence, weak topics, budget, recommended sessions, plan steps, explanation
Quiz	booking or group, topics, difficulty, questions with answer keys, status (draft or issued), issued time
QuizResult	quiz, student, answers, score, weak topics found

Privacy rule: ExamPlan and QuizResult are private to the student. Only counts and averages leave those records, and only through the dashboard.

Tool stack

Use each buildathon tool for one job, and show all three in your submission.

Tool	Job	Concrete use
DataCamp	Learn and analyze	Finish assigned courses on schedule; use the skills to clean seed data, chart price and demand patterns, and build the baseline forecast
Base44	Build the product	Entities, pages, filters, AI agents, quiz panel, dashboard
Hackhub	Present to judges	Problem, solution, data used, impact, demo link, roadmap
Meeting link provider (Zoom, Meet or Jitsi)	Online sessions	Store a link per booking; no custom video
Spreadsheet or notebook	Seed and analysis data	Prepare CSVs, then import them into Base44

Building the agents in Base44

- Write each agent as a clear prompt with a fixed input format and a fixed output format (JSON), so the UI can display results reliably.

- Keep decision rules (session counts, budget caps, free-first order) in app logic, and use the AI for language, extraction and explanation.
- Test every agent with 10 sample inputs, including bad ones (past exam date, $0 budget, unknown course).

Starter prompt for Base44 (paste, then refine page by page):

```
Build a web app called StudyLadder. Students pick their university and
location, then see two tabs: Free on-campus help and Paid tutors. Include
filters for subject, course code, price per session, availability,
qualifications, rating, and online or in person. Add tutor profiles, campus
service listings, ratings (only after a completed booking), study groups, an
Exam Prep Planner agent (inputs: course, exam date, confidence 1-5, weak
topics, availability, budget; output: recommended number of sessions, dates,
and free-first resource choices with an explanation), a Tutor Matcher agent
(plain-language request in, top 3 matches with reasons out), an online
session page where a tutor can generate an AI quiz, edit it, and issue it to
the student live, and an analytics dashboard with demand by course and week,
free vs paid usage, and money saved.
```

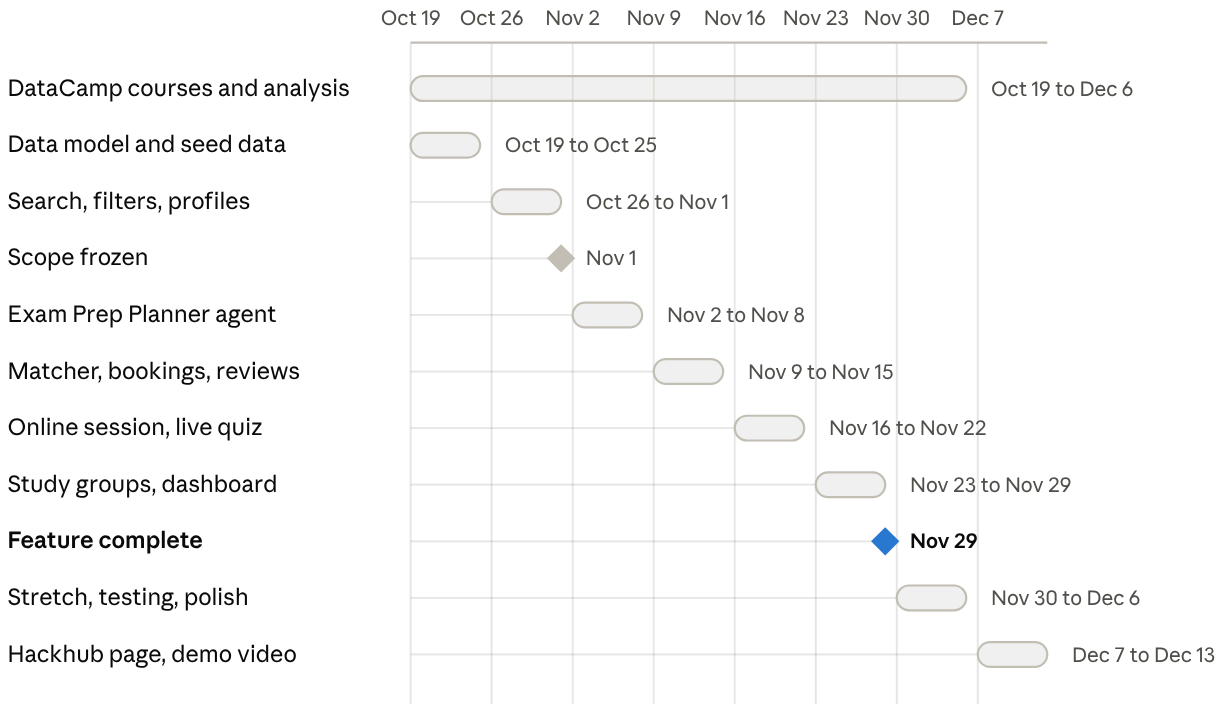
Check the buildathon rules for required datasets, technologies or submission formats before week 3, so nothing forces a late rebuild.

8-week execution plan

Two months is enough if you build the signature features first and stop adding features after week 6. This plan assumes a start on Monday, Oct 19, 2026; replace it with the buildathon's real deadline and count backward.

Feature complete lands at the end of week 6, leaving two weeks to test and submit

Eight weekly stages starting Oct 19, 2026; shift the bars to match the real deadline



8-week build timeline · 11 rows, 2 milestones

Each week ends with a check you can verify, so you know early if you are behind.

Week	Build focus	Data and DataCamp	Done when
1 (Oct 19)	Read buildathon rules; create the Base44 app; add entities from the data model; seed 3 campuses	Plan weekly DataCamp slots; list datasets	3 campuses and 15 sample tutors are in the app
2 (Oct 26)	Search with free and paid tabs, filters, tutor profiles, campus listings	Clean the seed CSVs	A student can find a tutor or service by course and price; scope is frozen
3 (Nov 2)	Exam Prep Planner: rules function plus AI explanation	Price distribution by subject	Plans work on 10 test inputs, including bad ones
4 (Nov 9)	Tutor Matcher, bookings, reviews after completed bookings	Event log for plans and bookings	Matcher returns 3 matches with reasons

Week	Build focus	Data and DataCamp	Done when
5 (Nov 16)	Online session page: generate, edit, issue and score a live quiz	Quiz result fields for analytics	Full quiz loop works end to end in about 2 minutes
6 (Nov 23)	Study groups and the 4-view dashboard	Demand by week, baseline forecast	Feature complete on Nov 29
7 (Nov 30)	1 or 2 stretch items; test with 5 real students; fix bugs	Verify every statistic and source	Persona journey finishes in under 3 minutes without help
8 (Dec 7)	Hackhub page, screenshots, backup video, demo rehearsal	Final impact estimate	Submitted with a day of buffer

If you fall behind: cut in this order: stretch items, group quiz, review summarizer, then reduce to 3 campuses. Never cut the Exam Prep Planner, the live quiz or the dashboard.

Demo script and Hackhub submission

A 3-minute demo that follows one student from panic to plan to live quiz.

Time	Scene	What the judges see
0:00 to 0:20	Hook	One stat or story: students pay for tutoring they could get free on campus
0:20 to 1:00	Exam Prep Planner	Persona enters Calc II, exam in 16 days, $0 to $30 budget; plan appears with a free-first ladder and explanations
1:00 to 1:25	Study group	Agent suggests a matching group; one-click RSVP
1:25 to 2:10	Live quiz	Tutor opens the session, generates a quiz, edits one question, issues it; student answers; weak topic flagged
2:10 to 2:40	Dashboard	Demand heatmap, free vs. paid share, money saved
2:40 to 3:00	Close	Roadmap and how it scales beyond pilot campuses

Hackhub project page checklist

- [ ] One-sentence pitch at the top
- [ ] Problem with a concrete statistic and its source
- [ ] Solution and the five pillars
- [ ] Screenshots of the planner, live quiz and dashboard
- [ ] Data used, what is simulated, and how privacy is handled
- [ ] How DataCamp learning shaped the analysis
- [ ] Impact estimate with its formula
- [ ] Roadmap and scaling plan
- [ ] Demo link and a backup recorded video

Record a backup demo video in week 8. Live demos fail at the worst moments.

Risks, scope cuts and safety

The biggest risk is scope, so decide now what gets cut and when.

Risk	Likelihood	Mitigation	Cut if late
Too many features, none polished	High	Freeze MVP list at week 2; no new features after week 6	Stretch table first
AI quiz gives wrong answers	Medium	Tutor reviews before issuing; show source topic; report button	Reduce to multiple choice only
Not enough real data	High	Seed 3 to 5 campuses; label sample data honestly	Reduce to 3 campuses
Session heuristic questioned	Medium	Call it a starting plan; cite spacing research you can verify	Keep rules table in pitch
Live demo fails	Medium	Backup video, pre-loaded demo account	None, always prepare
Base44 limits on a feature	Medium	Test hard features (live quiz, agents) in weeks 2 to 4, not week 7	Replace with simpler version

Risk	Likelihood	Mitigation	Cut if late
Falling behind on DataCamp courses	Medium	Schedule 4 to 5 hours per week, set weekly targets	Prioritize assigned courses

Safety and ethics

- Never present invented tutors or reviews as real; label sample data in the app and in the pitch.
- Verified badge is simulated in the demo; describe the real verification plan on the roadmap.
- No payment handling in the MVP, so no financial data to protect.
- Study group and tutor safety: campus email verification, public meeting places, report button.
- Tutors set their own rates; the app shows rates but does not guarantee outcomes.
- Say clearly that AI plans are suggestions and not a replacement for the instructor.

Path to a 10/10

The idea is strong on its own; execution and storytelling close the gap. Check each item before submitting. Judging criteria are not known to me, so confirm them in the buildathon guidelines and adjust this list.

Dimension	What a top score looks like	Where it lives in this plan
Real problem	Specific student pain with a sourced statistic	Vision, pitch hook
Originality	A clear point of view competitors don't share	Free-First Ladder, exam-date planning
AI that matters	Agents are central, explainable and guarded	AI agents, live quiz
Big Data use	Data drives decisions, with a visible pipeline	Dashboard, demand forecast, DataCamp analysis
Working product	Smooth demo of the full student journey	MVP list, backup video

Dimension	What a top score looks like	Where it lives in this plan
Impact	A number with a formula and honest assumptions	Money-saved metric
Inclusion	Designed for students who can least afford help	$0 budget path, mentor tag, accessibility filters
Scalability	Credible plan beyond pilot campuses	Roadmap, demand data for campuses
Ethics and privacy	Aggregates only, labeled sample data	Privacy rules, safety list

Final-week checks

- [ ] A stranger can complete the persona journey in under 3 minutes without help
- [ ] Every agent works on bad input without crashing
- [ ] Every number in the pitch has a source or a stated assumption
- [ ] Sample data is labeled everywhere it appears
- [ ] Backup video recorded and Hackhub page proofread by someone else